Now you're ready to jump into the scrubbing process itself. The first concept to tackle is how to keep the statistical properties the same when scrubbing data. What does that mean?

Replacement Values That Are Statistically Similar

The statistical properties of the data in tables, columns, and indexes all affect query execution performance.

Query execution plans are determined dynamically at execution time. PostgreSQL uses estimates and statistics about the data to estimate the cost of alternative plans. The query planner chooses the lowest cost plan that produces the fastest data retrieval. We'll get into details of query execution plan generation and plan selection later; for now, it's enough to know that table statistics are important to query planning.

What are some of the statistics collected by PostgreSQL?

As you've already learned, PostgreSQL keeps system catalog tables for internal bookkeeping. Statistics about the table rows that PostgreSQL keeps can be queried using the `pg_stats` System Catalog view.[2] The statistics include things like the most common values or how many rows have `NULL` values in their columns.

For the Rideshare database, take a look at the statistics captured for the `users` table by running the following query from psql. An example result row is shown below.

sql/table_statistics_users.sql

```
SELECT
  attname,
  n_distinct,
  most_common_vals
FROM pg_stats WHERE tablename = 'users'
AND attname = 'first_name';

-- # SELECT attname, n_distinct, most_common_vals FROM
-- pg_stats WHERE tablename = 'users' AND attname = 'first_name';
--
-- -[ RECORD 1 ]-
-- attname          | first_name
-- n_distinct       | 5122
-- most_common_vals | {Elroy,Maurice,Tristan,Dion,Ariel,Angel,Chi,
-- Chris,Frances,Luis,Mark,Ray,Fermin,Guadalupe,Logan,Perry,Royce,
-- Scott,Vernon,Carey,Carson,Cary,Dalton,Erin,Eugene,Frank,Glenn....
```

2. https://www.postgresql.org/docs/current/view-pg-stats.html